ML-Assisted Thermal Bias Prediction for High-Accuracy Accelerometers

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Team Alpha is interested in developing a machine learning model capable of predicting and compensating for thermal bias in high-accuracy accelerometers. Accelerometers embedded within Inertial Measurement Units (IMUs) experience changes in their output as internal components expand and contract with temperature fluctuations. This phenomenon, commonly referred to as **thermal bias** or **thermal drift**, reduces measurement accuracy and can negatively affect the performance of systems that depend on precise motion sensing.

IMUs, particularly those manufactured as integrated circuits (ICs), are widely used in drones, autonomous robots, smartphones, navigation systems, and aerospace applications to measure six degrees of freedom (6-DOF) motion. In these systems, even small measurement biases caused by changing temperatures can accumulate over time and significantly reduce navigation accuracy or control performance. The drone industry, specifically manufacturers of high speed, high precision drones like DJI, may disproportionately benefit from reduction of accelerometer error due to thermal drift. This industry depends on highly accurate inertial sensing and often employs calibration procedures to minimize thermal effects. Traditional calibration methods are typically sensor-specific and require individual characterization of every device. We propose that machine learning offers a more flexible alternative by learning generalized relationships between temperature measurements and thermal bias that can be applied across representative sensors rather than requiring extensive calibration for each individual unit. Inspiration for this approach was drawn from the articles under the “References” section, which detail the traditional calibration methods required to compensate for erroneous data due to thermal fluctuations.

Our hypothesis is that thermal bias follows predictable patterns that can be learned directly from temperature measurements collected throughout the accelerometer. Given sufficient highquality data, a machine learning model should be capable of accurately predicting the thermal bias experienced by the sensor, allowing that bias to be compensated for during operation.

During the preliminary research phase, we identified a dataset titled **"Temperature Characterization of a High Sensitivity Tri-Axial Accelerometer Through Multiple Thermometers"** by Iafolla et al. (2023). This publicly available dataset contains approximately one million observations in each of three CSV files. The data includes accelerometer outputs along the X, Y, and Z axes together with temperature measurements collected from eleven different sensor locations distributed throughout the accelerometer package and surrounding environment.

Unlike many sensor datasets that require construction of target variables, this dataset was specifically collected to characterize the influence of temperature on accelerometer measurements under controlled stationary conditions. Because the sensor remained stationary throughout testing, the measured accelerometer outputs represent thermal bias rather than actual motion. This makes the dataset well suited for supervised machine learning without requiring additional estimation or reconstruction of measurement error. The data dictionary for the dataset can be found below:

Variable Name	Description	Data Type	Role
z [m/s2]	Measured z-axis acceleration	float	target
x [m/s2]	Measured x-axis acceleration	float	target
y [m/s2]	Measured y-axis acceleration	float	target
T z [C]	Temperature of z-axis accelerometer element face	float	predictor
T x [C]	Temperature of x-axis accelerometer element face	float	predictor
T y [C]	Temperature of y-axis accelerometer element face	float	predictor

bottom x1 side [C]	Temperature of bottom accelerometer box (sensor 1)	float	predictor
x sensor face [C]	Temperature of accelerometer box x front face	float	predictor
y sensor face [C]	Temperature of accelerometer box y left face	float	predictor
bottom x2 side [C]	Temperature of bottom accelerometer box (sensor 2)	float	predictor
x elect face [C]	Temperature of accelerometer box x rear face	float	predictor
y ACQ face [C]	Temperature of accelerometer box y right face	float	predictor
top [C]	Temperature of accelerometer box top face	float	predictor
inside air [C]	Ambient temperature inside box	float	predictor
external [C]	Ambient temperature outside box	float	predictor
Altitude [m]	Altitude as measured via barometric pressure	float	predictor
Date	Date and time of sample measurement	Timestamp	Removed for analysis

The proposed modeling task is therefore to learn the relationship between the eleven temperature measurements and the corresponding thermal bias observed in the accelerometer outputs. Two modeling strategies will be explored:

- Develop three independent regression models to predict thermal bias along the X, Y, and Z axes individually.
- Develop a single multi-output regression model capable of simultaneously predicting thermal bias across all three axes.

The performance of both approaches will be compared to determine whether a unified model provides comparable or superior accuracy while simplifying deployment.

The preliminary analysis will begin with a Generalized Linear Model (GLM) as a baseline regression model to establish an interpretable benchmark and quantify the linear relationship between temperature measurements and thermal bias. More advanced machine learning approaches will then be evaluated, including a custom Deep Neural Network designed to capture nonlinear relationships between the temperature sensors and accelerometer bias.

Preprocessing will primarily focus on preparing the published dataset for model training rather than constructing new target variables. This includes:

- Handling any missing or invalid observations.
- Normalizing or standardizing temperature measurements.
- Exploring correlations among temperature sensors.
- Splitting the data into training, validation, and testing datasets.
- Evaluating whether feature engineering or dimensionality reduction improves predictive performance.

Model performance will be evaluated using standard regression metrics, including:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- Mean Absolute Percentage Error (MAPE), where appropriate
- Coefficient of Determination (R^2)

The final model will be compared against the baseline GLM to determine whether deep learning provides a meaningful improvement in predicting thermal bias while maintaining reasonable computational complexity. Success will be measured by the model's ability to accurately predict thermal bias on previously unseen test data and demonstrate a statistically significant improvement over the baseline model.

Achieving a production-ready thermal error elimination model in six weeks requires a strict schedule with actionable phases. In weeks 1 and 2, a comprehensive exploratory data analysis will be completed to understand the pre-existing features, as well as the influence of engineered features. Additionally, handling missing data via imputation will be performed as necessary. In weeks 3 and 4, the preliminary GLM model will be trained, tested, and an interpretability analysis will be performed to understand the relative effect of each feature on the target. This phase will also include the first steps in designing a deep neural network architecture to increase accuracy. Weeks 5 and 6 will be dedicated to finalizing the deep neural network and testing it against the GLM. Testing will span accuracy as well as real-time inference performance given the requirement for the model to perform at the sampling rate of the accelerometer. Finally, the results will be compiled, and a report of the results will be generated.

Python will serve as the primary programming language for this project. Key libraries will include NumPy, Pandas, Scikit-learn, TensorFlow, Matplotlib, and Seaborn for preprocessing, visualization, model development, and evaluation. GPU acceleration through NVIDIA CUDA will be used where available to reduce training time for deep learning models.

The expected outcome of this project is a generalized machine learning model capable of accurately predicting accelerometer thermal bias from distributed temperature measurements. Such a model has the potential to reduce calibration effort, improve inertial sensor accuracy, and support more reliable deployment of IMUs across robotics, aerospace, and industrial monitoring applications.

References:

Liu, D., Li, F., Zhao, G., Gao, L., Cai, B., Wu, X., Peng, M., Wu, W., & Tu, L. (2024). Compensation of temperature effects in force-balanced microelectromechanical system accelerometers. *Measurement*, *237*, Article 115126. https://doi.org/10.1016/j.measurement.2024.115126

Wang, Z., Cheng, X., & Du, J. (2020). Thermal Modeling and Calibration Method in Complex Temperature Field for Single-Axis Rotational Inertial Navigation System. *Sensors (Basel, Switzerland)*, *20*(2), 384. https://doi.org/10.3390/s20020384

Citations:

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